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|---|---|
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| SECTION: | S-8 |
| SUBJECT: | FULL STACK APPLICATION DEVELOPMENT |

# #SKILL - 8 Spring Boot – JPQL & Query Methods Module

***Tasks:***

**1. Create a Product entity with the following fields:**

**a. id**

**b. name**

**c. category**

**d. price**

**<u>Output:-</u>**

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LibraryCont... | LibraryAppAp... | Course.java | CourseServic... | Product.java

```java
package com.klu.skillab8.entity;

import jakarta.persistence.*;

@Entity
public class Product {

    @Id
    @GeneratedValue(strategy = GenerationType.IDENTITY)
    private Long id;

    private String name;
    private String category;
    private double price;

    public Product() {}

    public Product(String name, String category, double price) {
        this.name = name;
        this.category = category;
        this.price = price;
    }

    public Long getId() {
        return id;
    }

    public String getName() {
        return name;
    }

    public String getCategory() {
        return category;
    }

    public double getPrice() {
        return price;
    }

    public void setId(Long id) {
        this.id = id;
    }

    public void setName(String name) {
        this.name = name;
    }

    public void setCategory(String category) {
        this.category = category;
    }
```

**2. Create a ProductRepository and add derived query methods:**
**a. findByCategory(String category)**
**b. findByPriceBetween(double min, double max)**

**Output:-**

LibraryCont... | LibraryAppAp... | Course.java | Product.java | ProductRepos...

```java
package com.klu.skillab8.repository;

import org.springframework.data.jpa.repository.JpaRepository;

public interface ProductRepository extends JpaRepository<Product, Long> {

    // Derived Queries
    Show AOT-generated Implementation, Query, etc...
    List<Product> findByCategory(String category);

    Show AOT-generated Implementation, Query, etc...
    List<Product> findByPriceBetween(double min, double max);

    // JPQL Queries

    // a. Sorting products by price
    Show AOT-generated Implementation, Query, etc...
    @Query("SELECT p FROM Product p ORDER BY p.price ASC")
    List<Product> sortProductsByPrice();

    // b. Fetch products above a price value
    Show AOT-generated Implementation, Query, etc...
    @Query("SELECT p FROM Product p WHERE p.price > :price")
    List<Product> findExpensiveProducts(double price);

    // c. Fetch products by category
    Show AOT-generated Implementation, Query, etc...
    @Query("SELECT p FROM Product p WHERE p.category = :category")
    List<Product> findProductsByCategory(String category);
}
```

**3. Write JPQL queries using @Query for:**
**a. Sorting products by price**
**b. Fetching products above a price value**
**c. Fetching products by category** **Output:-**

```java
public interface ProductRepository extends JpaRepository<Product, Long> {

    // Derived Queries
    Show AOT-generated Implementation, Query, etc...
    List<Product> findByCategory(String category);

    Show AOT-generated Implementation, Query, etc...
    List<Product> findByPriceBetween(double min, double max);

    // JPQL Queries

    // a. Sorting products by price
    Show AOT-generated Implementation, Query, etc...
    @Query("SELECT p FROM Product p ORDER BY p.price ASC")
    List<Product> sortProductsByPrice();

    // b. Fetch products above a price value
    Show AOT-generated Implementation, Query, etc...
    @Query("SELECT p FROM Product p WHERE p.price > :price")
    List<Product> findExpensiveProducts(double price);

    // c. Fetch products by category
    Show AOT-generated Implementation, Query, etc...
    @Query("SELECT p FROM Product p WHERE p.category = :category")
    List<Product> findProductsByCategory(String category);
}
```

**4. Create REST controller endpoints to test query methods:**
**a. /products/category/{category}**
**b. /products/filter?min=&max=**
**c. /products/sorted d. /products/expensive/{price}**

**Output:-**

```java
import org.springframework.beans.factory.annotation.Autowired;

@RestController
@RequestMapping("/products")
public class ProductController {

    @Autowired
    ProductRepository productRepository;

    // a. /products/category/{category}
    @GetMapping("/category/{category}")
    public List<Product> getByCategory(@PathVariable String category) {
        return productRepository.findByCategory(category);
    }

    // b. /products/filter?min=&max=
    @GetMapping("/filter")
    public List<Product> filterByPrice(@RequestParam double min,
                                       @RequestParam double max) {
        return productRepository.findByPriceBetween(min, max);
    }

    // c. /products/sorted
    @GetMapping("/sorted")
    public List<Product> getSortedProducts() {
        return productRepository.sortProductsByPrice();
    }

    // d. /products/expensive/{price}
    @GetMapping("/expensive/{price}")
    public List<Product> getExpensiveProducts(@PathVariable double price) {
        return productRepository.findExpensiveProducts(price);
    }
}
```

**5. Insert sample product records and test using Postman.**
**Output:-**

GET http://localhost:8080/products/category/Electronics

Docs Params Auth Headers (8) Body Scripts Tests Set

Query Params

| Key | Value |
|---|---|
| Key | Value |

Body 200 OK

{} JSON Preview Visualize


```
1  [
2      {
3          "name": "Laptop",
4          "category": "Electronics",
5          "price": 75000.0,
6          "id": 1
7      },
8      {
9          "name": "Mobile",
10         "category": "Electronics",
11         "price": 30000.0,
12         "id": 2
13     },
14     {
15         "name": "Headphones",
16         "category": "Electronics",
17         "price": 2000.0,
18         "id": 7
```

GET http://localhost:8080/products/filter?min=1000&max=5000

Docs | Params | Auth | Headers (8) | Body | Scripts | Tests | Settings

Query Params

| Key | Value | Descript |
|---|---|---|
| min | 1000 | |
| max | 5000 | |
| Key | Value | Descript |

Body 200 OK 72

JSON | Preview | Visualize


```
[
    {
        "name": "Shoes",
        "category": "Fashion",
        "price": 2500.0,
        "id": 3
    },
    {
        "name": "T-Shirt",
        "category": "Fashion",
        "price": 1200.0,
        "id": 4
    },
    {
        "name": "Headphones",
        "category": "Electronics",
        "price": 2000.0,
        "id": 7
```


GET http://localhost:8080/products/sorted

Docs | Params | Auth | Headers (8) | Body | Scripts | Tests | S

Query Params

| Key | Value |
|---|---|
| Key | Value |

Body 200

JSON | Preview | Visualize


```
[
    {
        "name": "T-Shirt",
        "category": "Fashion",
        "price": 1200.0,
        "id": 4
    },
    {
        "name": "Headphones",
        "category": "Electronics",
        "price": 2000.0,
        "id": 7
    },
    {
        "name": "Shoes",
        "category": "Fashion",
        "price": 2500.0,
        "id": 3
    },
    {
        "name": "Mobile",
```

GET http://localhost:8080/products/expensive/20000

Docs Params Auth Headers (8) Body Scripts Tests Se

Query Params

| | Key | Value |
|---|---|---|
| | Key | Value |

Body 200

{ } JSON Preview Visualize


```
[
    {
        "name": "Laptop",
        "category": "Electronics",
        "price": 75000.0,
        "id": 1
    },
    {
        "name": "Mobile",
        "category": "Electronics",
        "price": 30000.0,
        "id": 2
    },
    {
        "name": "Refrigerator",
        "category": "Appliances",
        "price": 45000.0,
        "id": 5
    },
    {
        "name": "Washing Machine",
```


7. Push the Spring Boot project to GitHub.

PulluruSathvik / skillab8

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skillab8 initial commit 3 hours ago

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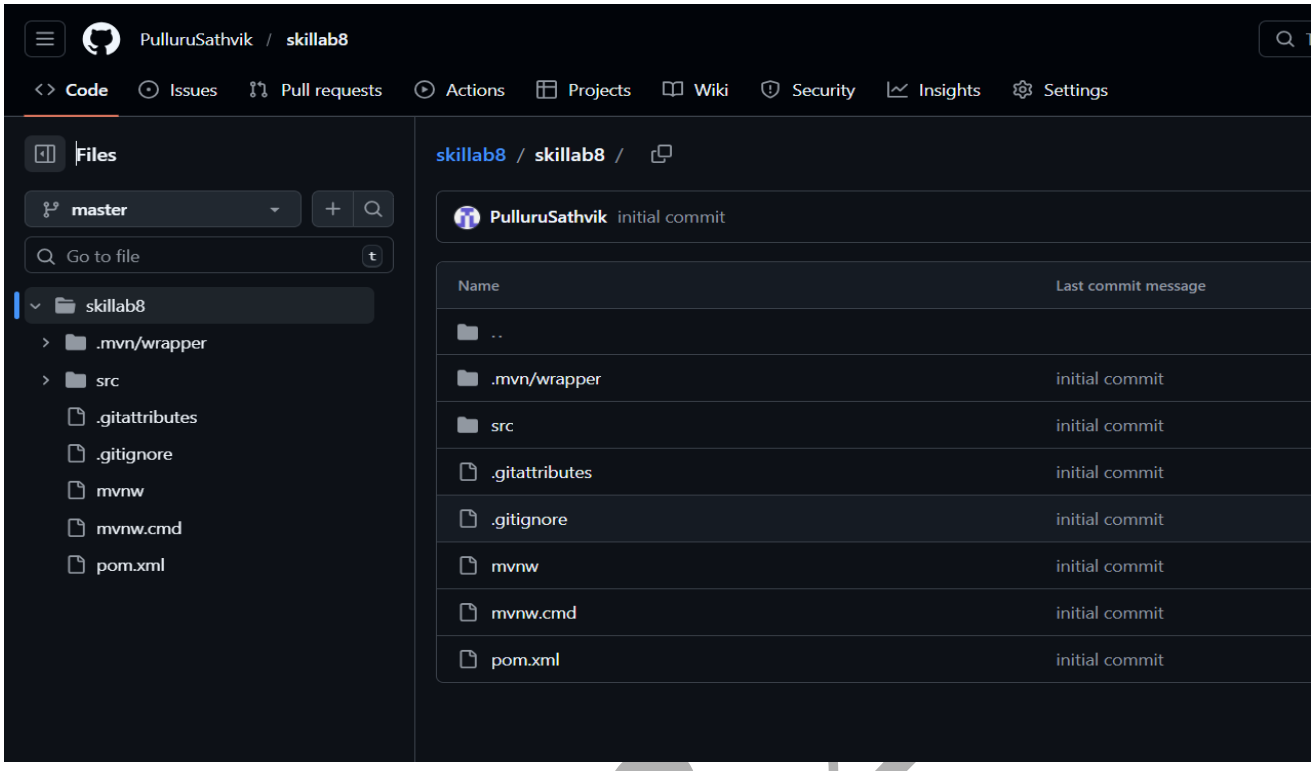


GITHUB REPO LINK :- https://github.com/PulluruSathvik/skillab8

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